

A disproof of Conjecture 00000001951

$\text{Out}(F_n)$ has a normal subgroup of index 2, so the smallest index of a proper subgroup is
not $2^n \cdot C(n, 2)$

1 The conjecture

Conjecture 00000001951 reads (English version; the Chinese version states the same conjecture, also for $\text{Out}(F_n)$):

Definition: The congruence subgroup property (CSP) of $\text{Out}(F_n)$ is the coincidence of its congruence subgroups with all finite-index subgroups. *Conjecture:* $\text{Out}(F_n)$ ($n \geq 3$) fails CSP (its smallest-index proper subgroup is an automorphism kernel); the smallest index of a proper subgroup is $2^n \cdot C(n, 2)$.

Reading. Here $C(n, 2) = \binom{n}{2}$. The conjecture is the conjunction of two clauses, for every $n \geq 3$:

- (a) $\text{Out}(F_n)$ fails CSP (with the parenthetical remark that its smallest-index proper subgroup is an automorphism kernel);
- (b) the smallest index of a proper subgroup of $\text{Out}(F_n)$ is $2^n \cdot C(n, 2)$.

We refute clause (b) for every $n \geq 3$ (indeed for every $n \geq 2$), which refutes the conjunction. We do **not** address clause (a): whether $\text{Out}(F_n)$, $n \geq 3$, has the congruence subgroup property is open, and the refutation below does not depend on it in any way.

“Proper subgroup” is read as a proper subgroup of finite index ($H \neq \text{Out}(F_n)$ and $[\text{Out}(F_n) : H] < \infty$), since only these have an index that is a positive integer. (Admitting infinite-index subgroups, with index ∞ , would not change a minimum that is attained at a finite value.) The index-2 subgroup we exhibit is moreover *normal*, so the reading “smallest index of a proper normal subgroup” is refuted as well. The statement is about $\text{Out}(F_n)$ in both the English and the Chinese versions, so we do not treat the special outer automorphism group $\text{SOut}(F_n)$ or any other subgroup in place of $\text{Out}(F_n)$. (The subgroup K constructed below is the one usually denoted $\text{SOut}(F_n)$, the image of $\text{SAut}(F_n)$ in $\text{Out}(F_n)$; this identification is not used.)

2 Conventions

These are the standard definitions, and they are exactly the ones used in the Lean file.

- F_n is the free group on $x_0, \dots, x_{n-1}$ (Lean `FreeGroup (Fin n)`); $\text{Aut}(F_n)$ is its group of automorphisms (Lean `MulAut (F n)`), with multiplication $\varphi\psi = \varphi \circ \psi$.
- $\text{Inn}(F_n)$ is the range of the homomorphism $g \mapsto \text{conj}_g$, $\text{conj}_g(x) = gxg^{-1}$ (Lean `MulAut.conj`); it is normal (Lemma 1), and $\text{Out}(F_n) = \text{Aut}(F_n)/\text{Inn}(F_n)$.

- The abelianization $F_n \rightarrow F_n^{ab} \cong \mathbb{Z}^n$ is realised concretely as the exponent-sum homomorphism $\text{vec} : F_n \rightarrow \mathbb{Z}^n$, $x_i \mapsto e_i$ (the i -th standard basis vector), so $\text{vec}(xy) = \text{vec } x + \text{vec } y$ and $\text{vec}(x^{-1}) = -\text{vec } x$.
- For $\varphi \in \text{Aut}(F_n)$, $\text{mat } \varphi$ is the integer $n \times n$ matrix whose j -th column is $\text{vec}(\varphi(x_j))$, i.e. $(\text{mat } \varphi)_{ij} = \text{vec}(\varphi(x_j))_i$.
- Mathlib's `Subgroup.index` is 0 for a subgroup of infinite index. All our statements about proper subgroups require finite index (`FiniteIndex`, i.e. `index` $\neq 0$), so nothing relies on that convention. Likewise `sInf` of a set of natural numbers is its least element when the set is nonempty, which is the case here.

3 The proof

Lemma 1 (Lean `Inn.normal`). *$\text{Inn}(F_n)$ is a normal subgroup of $\text{Aut}(F_n)$.*

Proof. For $\psi \in \text{Aut}(F_n)$ and $g \in F_n$: $\psi \text{ conj}_g \psi^{-1}(x) = \psi(g \psi^{-1}(x) g^{-1}) = \psi(g) x \psi(g)^{-1}$, so $\psi \text{ conj}_g \psi^{-1} = \text{conj}_{\psi(g)} \in \text{Inn}(F_n)$. $\square$

Lemma 2 (Lean `vec_apply`). *For $\varphi \in \text{Aut}(F_n)$ and $w \in F_n$: $\text{vec}(\varphi(w)) = \text{mat}(\varphi) \text{vec}(w)$.*

Proof. Induction on words. For $w = 1$ both sides are 0. For $w = x_j$, $\text{mat}(\varphi) e_j$ is the j -th column $\text{vec}(\varphi(x_j))$. For $w = x_j^{-1}$, $\text{vec}(\varphi(x_j^{-1})) = -\text{vec}(\varphi(x_j)) = -\text{mat}(\varphi) e_j = \text{mat}(\varphi) \text{vec}(x_j^{-1})$. For $w = xy$, both sides are additive in w (φ and vec are homomorphisms, matrix-vector multiplication is linear). $\square$

Lemma 3 (Lean `mat_one`, `mat_mul`). *$\text{mat}(1) = I$ and $\text{mat}(\varphi\psi) = \text{mat}(\varphi) \text{mat}(\psi)$.*

Proof. The j -th column of $\text{mat}(1)$ is $\text{vec}(x_j) = e_j$. The j -th column of $\text{mat}(\varphi\psi)$ is $\text{vec}(\varphi(\psi(x_j))) = \text{mat}(\varphi) \text{vec}(\psi(x_j))$ by Lemma 2, which is $\text{mat}(\varphi)$ times the j -th column of $\text{mat}(\psi)$. $\square$

Hence $\text{mat} : \text{Aut}(F_n) \rightarrow M_n(\mathbb{Z})$ is a monoid homomorphism (`matHom`), and so is $\det \circ \text{mat} : \text{Aut}(F_n) \rightarrow (\mathbb{Z}, \cdot)$. Since $\text{Aut}(F_n)$ is a group, its image consists of units, giving a group homomorphism $\det\text{Hom} : \text{Aut}(F_n) \rightarrow \mathbb{Z}^\times = \{\pm 1\}$ with value $\det(\text{mat } \varphi)$ (Lean `detHom` via `MonoidHom.toHomUnits`, and `detHom_val`).

Lemma 4 (Lean `mat_conj`, `Inn_le_ker`). *$\text{mat}(\text{conj}_g) = I$ for every $g \in F_n$. Hence $\text{Inn}(F_n) \leq \ker(\det\text{Hom})$, and $\det\text{Hom}$ factors through a homomorphism $\text{outDet} : \text{Out}(F_n) \rightarrow \mathbb{Z}^\times$, $[\varphi] \mapsto \det(\text{mat } \varphi)$ (Lean `outDet`, via `QuotientGroup.lift`).*

Proof. $\text{vec}(gx_jg^{-1}) = \text{vec } g + e_j - \text{vec } g = e_j$, so every column of $\text{mat}(\text{conj}_g)$ is the corresponding column of I , and $\det I = 1$. $\square$

Lemma 5 (Lean `flip`, `mat_flip`, `det_flip`). *Fix i_0 with $0 \leq i_0 < n$. The endomorphism f_{i_0} of F_n with $x_{i_0} \mapsto x_{i_0}^{-1}$ and $x_i \mapsto x_i$ ($i \neq i_0$) is an involution, hence an automorphism; $\text{mat}(f_{i_0})$ is the diagonal matrix with entry -1 at i_0 and 1 elsewhere, and $\det \text{mat}(f_{i_0}) = -1$.*

Proof. $f_{i_0} \circ f_{i_0}$ fixes every generator, so it is the identity, and f_{i_0} is its own inverse. The j -th column of $\text{mat}(f_{i_0})$ is $\text{vec}(x_{i_0}^{-1}) = -e_{i_0}$ for $j = i_0$ and $\text{vec}(x_j) = e_j$ otherwise. The determinant of a diagonal matrix is the product of its diagonal entries, here -1 . $\square$

Lemma 6 (Lean `outDet_surjective`). *If $n \geq 1$, then $\text{outDet} : \text{Out}(F_n) \rightarrow \mathbb{Z}^\times$ is surjective.*

Proof. $\mathbb{Z}^\times = \{1, -1\}$. We have $\text{outDet}(1) = 1$, and since $n \geq 1$ the generator x_0 exists, so $\text{outDet}([f_0]) = \det \text{mat}(f_0) = -1$ by Lemma 5. $\square$

Theorem 7 (Lean `K_index, exists_normal_index_two`). *Let $n \geq 1$ and $K = \ker(\text{outDet}) \leq \text{Out}(F_n)$. Then K is a proper normal subgroup of $\text{Out}(F_n)$ of index exactly 2.*

Proof. K is normal as a kernel. By the first isomorphism theorem and Lemma 6, $\text{Out}(F_n)/K \cong \mathbb{Z}^\times$, so $[\text{Out}(F_n) : K] = |\mathbb{Z}^\times| = 2$. Since the whole group has index $1 \neq 2$, $K \neq \text{Out}(F_n)$. $\square$

Theorem 8 (Lean `sInf_properIndices`). *For $n \geq 1$ let $\text{properIndices}(n) = \{[\text{Out}(F_n) : H] : H \neq \text{Out}(F_n), [\text{Out}(F_n) : H] < \infty\}$. Then its least element is 2.*

Proof. 2 belongs to the set by Theorem 7. Conversely, if H is proper of finite index, then its index is a positive integer, and it is not 1 because a subgroup of index 1 is the whole group; so the index is at least 2. $\square$

Lemma 9 (Lean `two_lt`). *For $n \geq 2$: $2 < 2^n \cdot C(n, 2)$.*

Proof. $2^n \geq 2^2 = 4$ and $C(n, 2) \geq 1$, so $2^n \cdot C(n, 2) \geq 4 > 2$. $\square$

Theorem 10 (Lean `smallest_index_ne, not_all_index_ge`). *For every $n \geq 3$, the smallest index of a proper (finite-index) subgroup of $\text{Out}(F_n)$ is 2, which is different from $2^n \cdot C(n, 2)$. Equivalently, not every proper finite-index subgroup of $\text{Out}(F_n)$ has index at least $2^n \cdot C(n, 2)$. Hence clause (b), and with it the conjecture, is false.*

Proof. Theorem 8 gives the minimum 2, and Lemma 9 gives $2 < 2^n \cdot C(n, 2)$. The subgroup K of Theorem 7 is a proper finite-index subgroup of index $2 < 2^n \cdot C(n, 2)$. $\square$

Remark 11. For $n = 3$ the conjectured value is $2^3 \cdot C(3, 2) = 8 \cdot 3 = 24$, whereas $\text{Out}(F_3)$ has the proper normal subgroup K of index 2. Since K is normal and every proper finite-index subgroup has index at least 2, the minimum is also 2 under the reading “proper normal subgroup”. The argument uses only the action on $F_n^{ab} \cong \mathbb{Z}^n$ and is valid for every $n \geq 1$; the hypothesis $n \geq 2$ is needed only for $2 < 2^n \cdot C(n, 2)$ (for $n = 1$ the right side is 0).

4 Lean 4 formalization

The directory `lean/` contains a Lean 4 project (Lean `v4.33.1`, Mathlib `v4.33.1`). The single source file `Conjecture1951/Basic.lean` (namespace `C1951`) defines `F n := FreeGroup (Fin n)`, `Inn n := (MulAut.conj).range`, `Out n := MulAut (F n) / Inn n` (quotient group), the exponent-sum map `vec`, the matrix `mat`, the characters `detHom` and `outDet`, the automorphism `flip`, `K n := (outDet n).ker`, and

`properIndices n := {k | ∃ H : Subgroup (Out n), H ≠ ⊤ ∧ H.FiniteIndex ∧ H.index = k}`.

Main results:

Lean name	Statement
<code>Inn.normal</code>	<code>instance: (Inn n).Normal</code>
<code>vec_apply</code>	$\text{vec}(\varphi w) = \text{mat } \varphi \cdot \text{vec } w$ (matrix-vector product <code>mulVec</code>)
<code>mat_mul</code>	$\text{mat}(\varphi\psi) = \text{mat } \varphi \cdot \text{mat } \psi$ (and <code>mat_one</code> : $\text{mat } 1 = 1$)
<code>mat_conj</code>	$\text{mat}(\text{MulAut.conj } g) = 1$
<code>outDet</code>	definition: $\text{Out } n \rightarrow^* \mathbb{Z}^\times$, the lift of <code>detHom</code> through <code>Inn_</code> <code>le_ker</code> : $\text{Inn } n \leq (\text{detHom } n).\text{ker}$
<code>det_flip</code>	$\text{det}(\text{mat}(\text{flip } i_0)) = -1$ (via <code>mat_flip</code> : diagonal, -1 at i_0 , 1 elsewhere)
<code>outDet_surjective</code>	$1 \leq n \rightarrow \text{Function.Surjective } (\text{outDet } n)$
<code>K_index</code>	$1 \leq n \rightarrow (\text{K } n).\text{index} = 2$
<code>exists_normal_index_two</code>	$1 \leq n \rightarrow \exists H : \text{Subgroup } (\text{Out } n), H.\text{Normal} \wedge H \neq \top \wedge H.\text{index} = 2$
<code>sInf_properIndices</code>	$1 \leq n \rightarrow \text{sInf } (\text{properIndices } n) = 2$
<code>two_lt</code>	$2 \leq n \rightarrow 2 < 2^n \cdot n.\text{choose } 2$
<code>smallest_index_ne</code>	$3 \leq n \rightarrow \text{sInf } (\text{properIndices } n) \neq 2^n \cdot n.\text{choose } 2$
<code>not_all_index_ge</code>	$3 \leq n \rightarrow \neg \forall H : \text{Subgroup } (\text{Out } n), H \neq \top \rightarrow H.\text{FiniteIndex} \rightarrow 2^n \cdot n.\text{choose } 2 \leq H.\text{index}$

There is no sorry, no `native_decide` and no added axiom. `#print axioms` for `smallest_index_ne`, `not_all_index_ge`, `sInf_properIndices` and `exists_normal_index_two` reports only `propext`, `Classical.choice` and `Quot.sound`. To check:

```
cd lean
lake exe cache get
lake build
```